

# Memória

sistemas de numeração

0 1

0 1 2 3 4 5 6 7 8 9

0 1 2 3 4 5 6 7 8 9 a b c d e f

0 1 2 3 4 5 6 7 8 9 A B C D E F

hexadecimal

base 16

$16^1$

$16^0$

#

#

16

1

#

#

16 1

00

16 1

01

16 1

02

16 1

03

16 1

04

16 1

05

16 1

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16 1

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08

16 1

09

16 1

0A

16 1

0B

16 1

0C

16 1

ØD

16 1

0E

16 1

OF

16 1

10

16 1

11

16 1

12

16 1



16 1

FF

16 1

FF

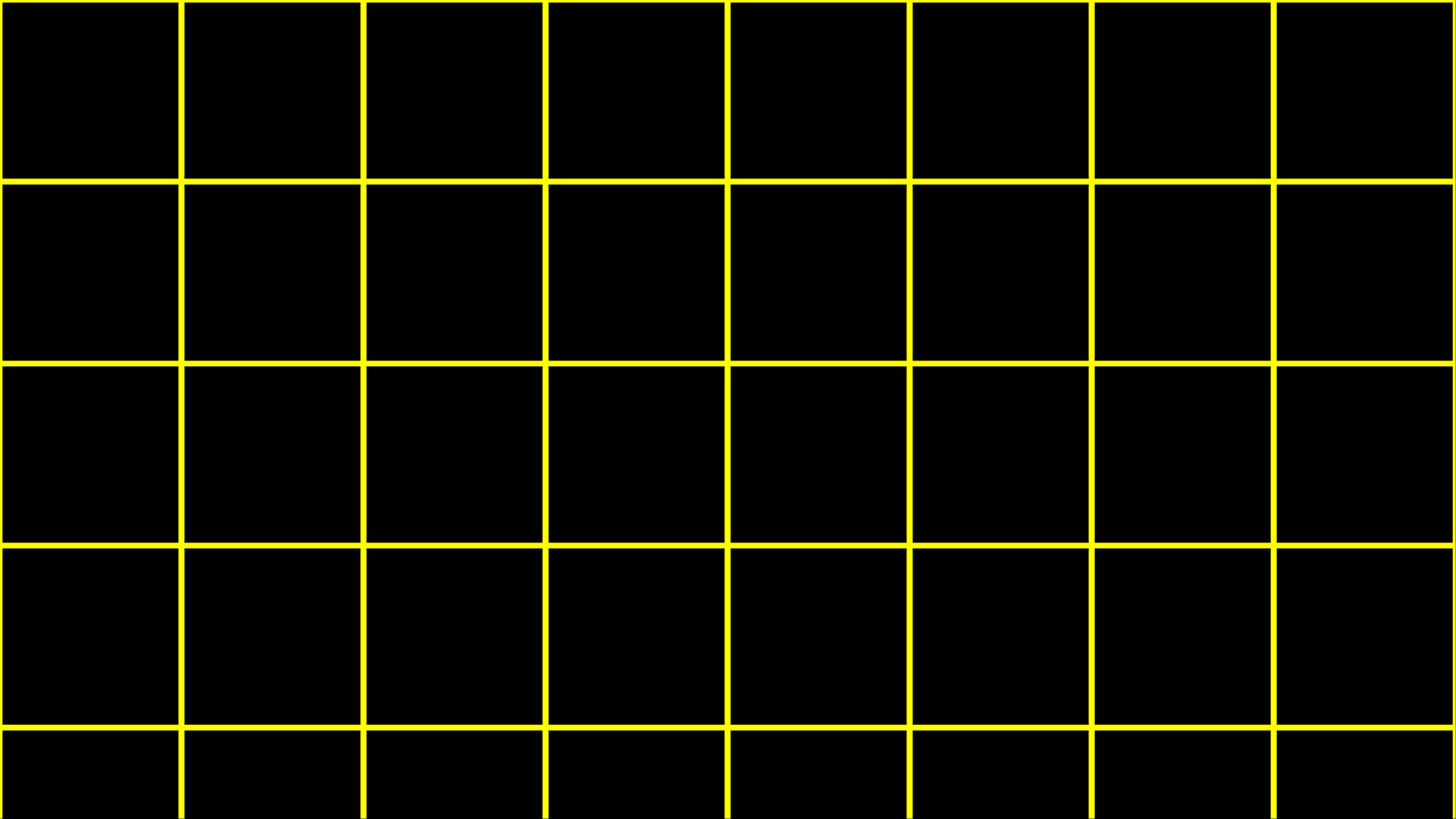
$$16 \times F + 1 \times F = 255$$

FF

F F

1111

1111



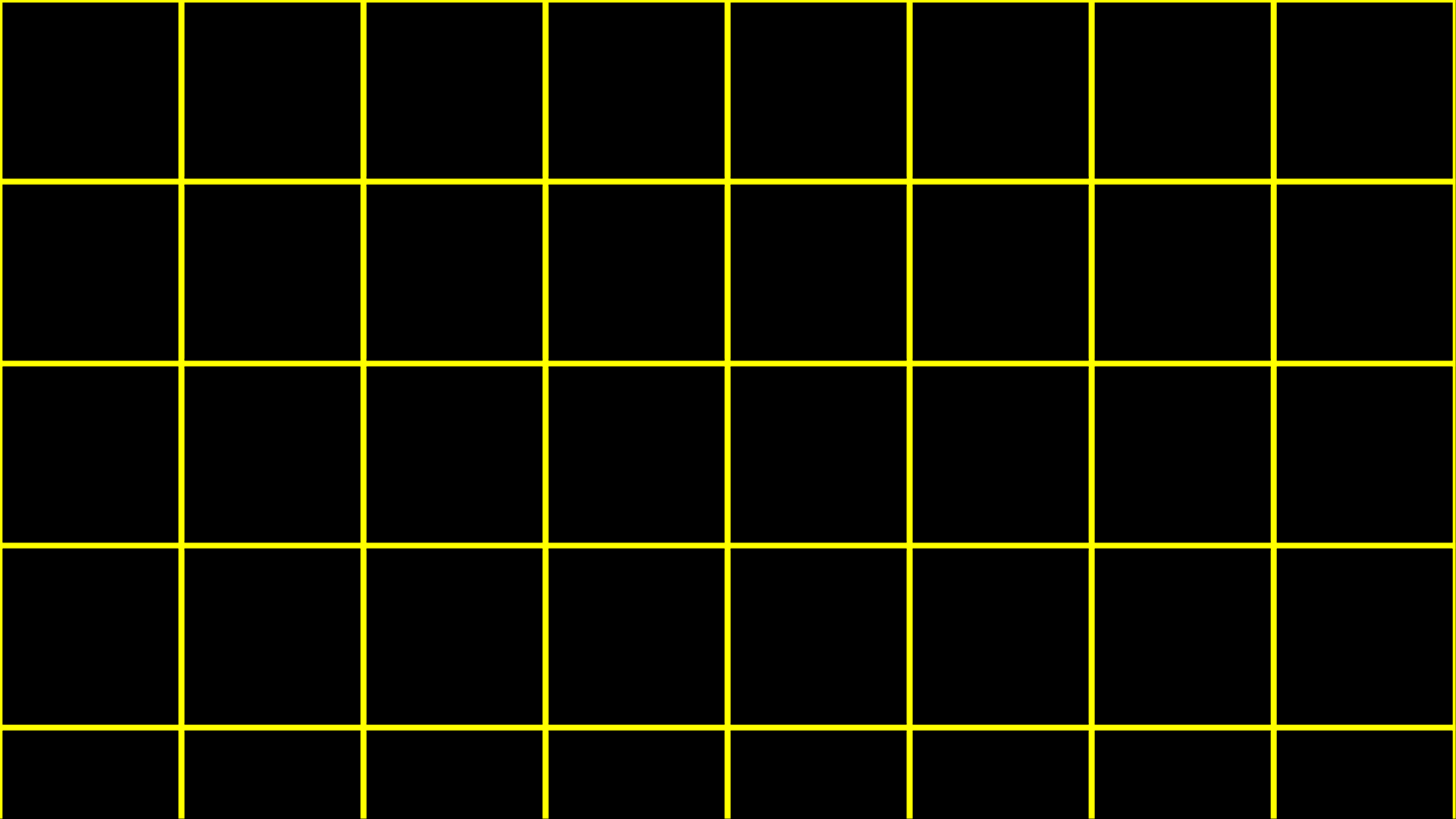
|   |   |    |    |    |    |    |    |
|---|---|----|----|----|----|----|----|
| 0 | 1 | 2  | 3  | 4  | 5  | 6  | 7  |
| 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 |
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|---|---|---|---|---|---|---|---|
| 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| 8 | 9 | A | B | C | D | E | F |
|   |   |   |   |   |   |   |   |
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|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 0  | 1  | 2  | 3  | 4  | 5  | 6  | 7  |
| 8  | 9  | A  | B  | C  | D  | E  | F  |
| 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 |
| 18 | 19 | 1A | 1B | 1C | 1D | 1E | 1F |
|    |    |    |    |    |    |    |    |

|      |      |      |      |      |      |      |      |
|------|------|------|------|------|------|------|------|
| 0x0  | 0x1  | 0x2  | 0x3  | 0x4  | 0x5  | 0x6  | 0x7  |
| 0x8  | 0x9  | 0xA  | 0xB  | 0xC  | 0xD  | 0xE  | 0xF  |
| 0x10 | 0x11 | 0x12 | 0x13 | 0x14 | 0x15 | 0x16 | 0x17 |
| 0x18 | 0x19 | 0x1A | 0x1B | 0x1C | 0x1D | 0x1E | 0x1F |
|      |      |      |      |      |      |      |      |

```
int n = 50;
```



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|  |  |  |  | 50 |  |  |  |
|  |  |  |  | n  |  |  |  |
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|  |  |  |  | 50    |  |  |  |
|  |  |  |  | 0x123 |  |  |  |
|  |  |  |  |       |  |  |  |

&

\*

%p

ponteiros

```
int n = 50;
```

```
int *p = &n;
```

```
int n = 50;
```

```
int *p = &n;
```

```
int n = 50;
```

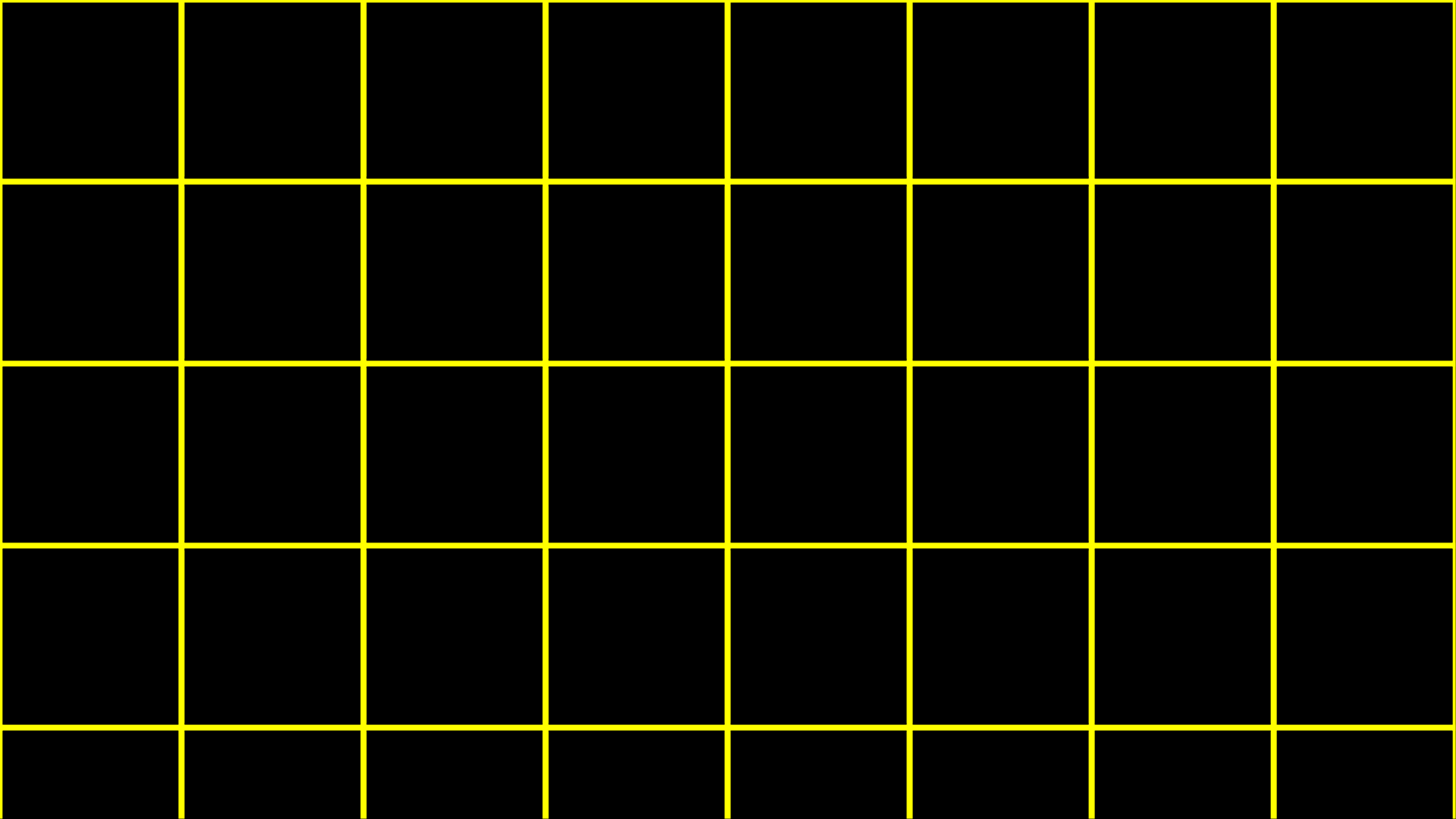
```
int* p = &n;
```

```
int n = 50;
```

```
int * p = &n;
```

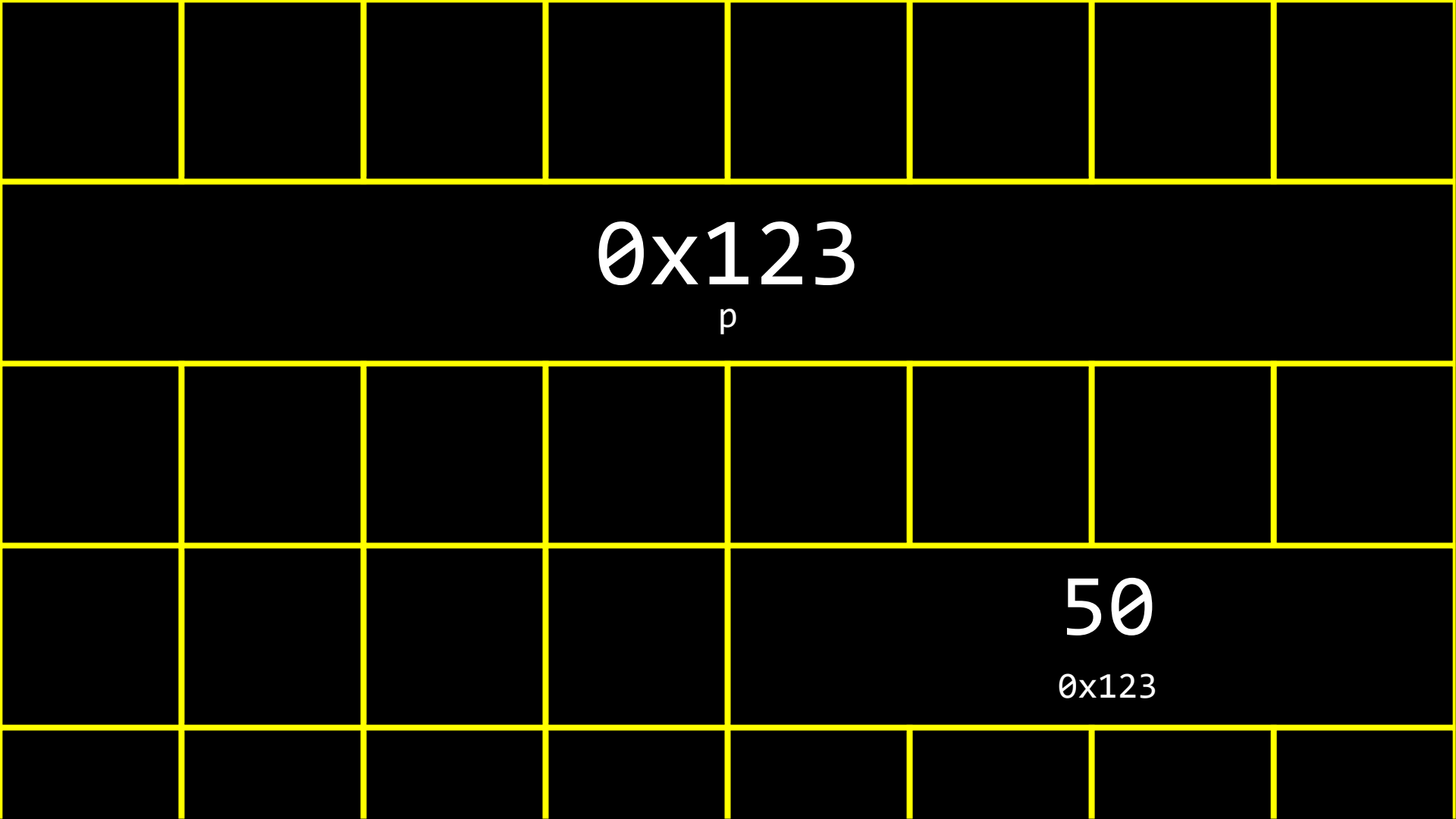
```
int n = 50;
```

```
int* p = &n;
```



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|  |  |  |  |    |  |  |  |
|  |  |  |  |    |  |  |  |
|  |  |  |  | 50 |  |  |  |
|  |  |  |  | n  |  |  |  |
|  |  |  |  |    |  |  |  |

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|  |  |  |  |       |  |  |  |
|  |  |  |  | 50    |  |  |  |
|  |  |  |  | 0x123 |  |  |  |
|  |  |  |  |       |  |  |  |



0x123

p

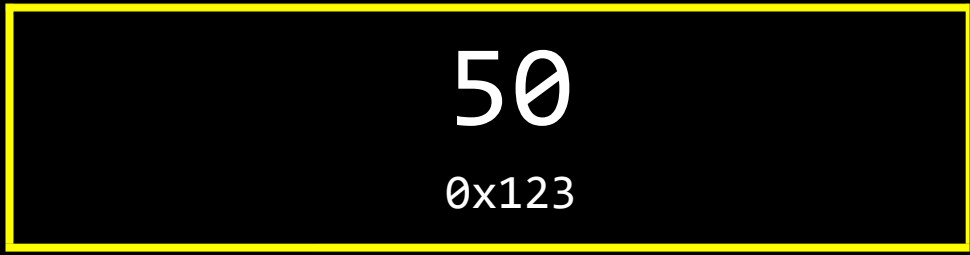
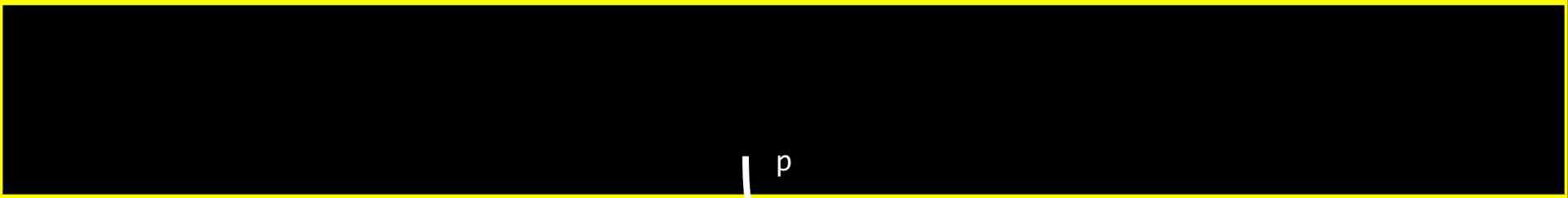
50

0x123

0x123  
p

50

0x123



```
string nome = "Davi Alves";
```

```
string numero = "6940";
```

```
typedef struct  
{  
    string nome;  
    string numero;  
} aluno;
```

```
typedef struct  
{  
    string nome;  
    string numero;  
} aluno;
```

```
typedef struct  
{  
    string nome;  
    string numero;  
} aluno;
```

```
typedef struct  
{  
    string nome;  
    string numero;  
} aluno;
```

typedef

```
typedef int
```

```
typedef int inteiro;
```

```
typedef float decimal;
```

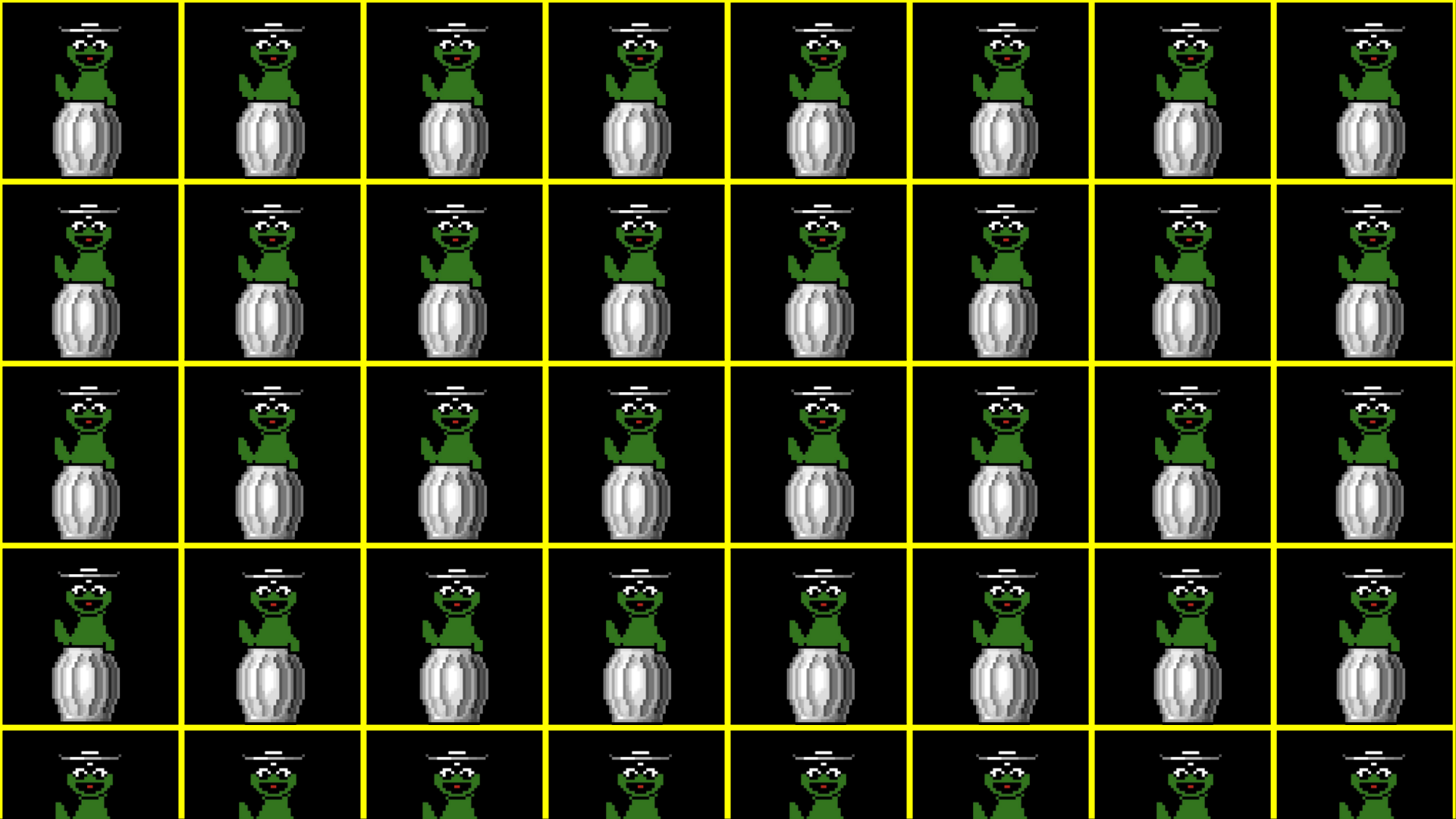
Valores lixo

```
int main(void)
{
    int x = 7;
    int* y;
    int* z = &x;

    *z = 42;
    *y = 13;

    y = z;

    *y = 13;
}
```



```
int main(void)
{
    int x = 7;
    int* y;
    int* z = &x;

    *z = 42;
    *y = 13;

    y = z;

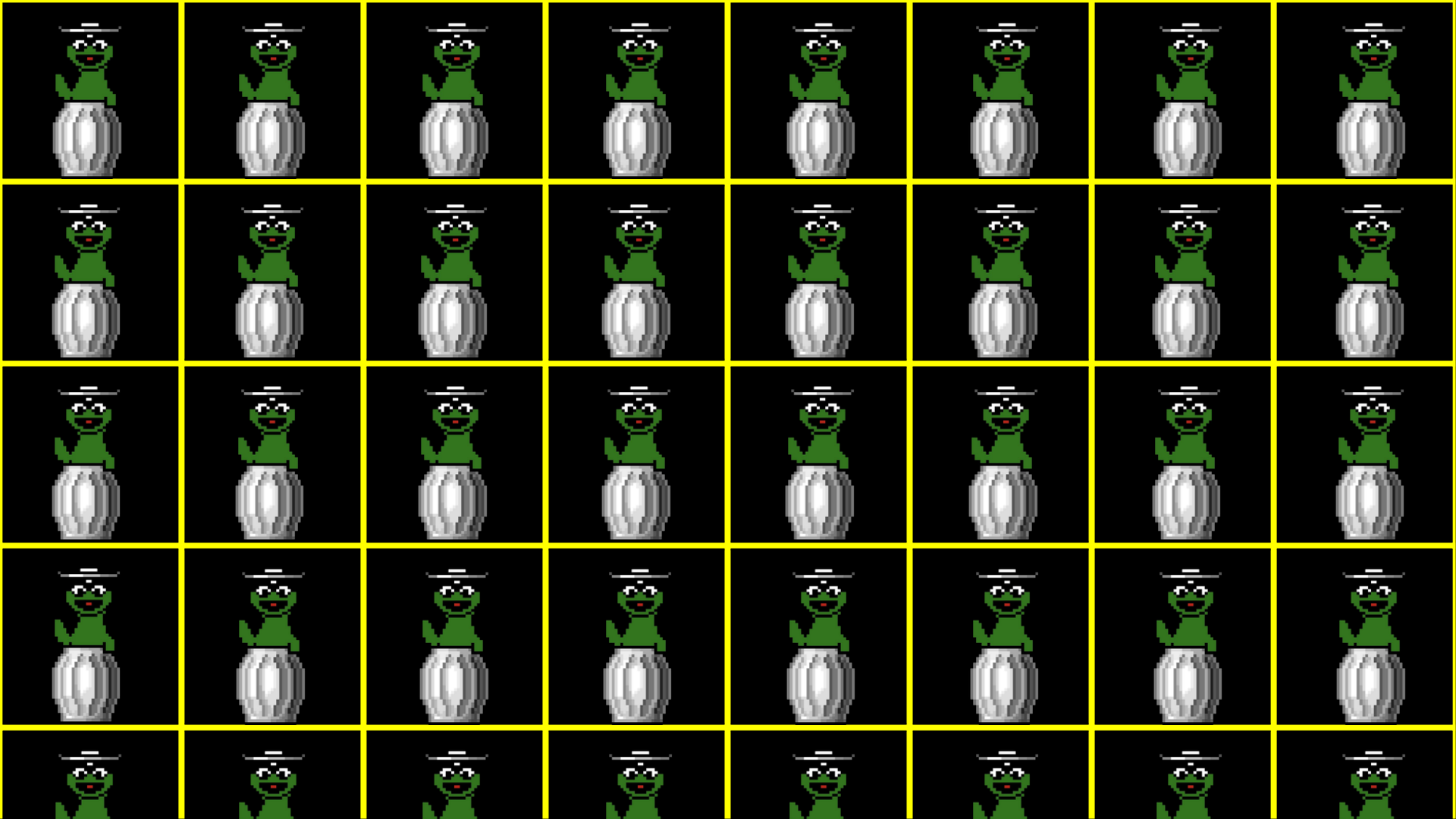
    *y = 13;
}
```

```
int main(void)
{
    int x = 7;
    int* y;
    int* z = &x;

    *x = 42;
    *y = 13;

    y = x;

    *y = 13;
}
```





???

y [0x234]

# 0x123

z [0x321]



7

x [0x123]



```
int main(void)
{
    int x = 7;
    int* y;
    int* z = &x;

    *z = 42;
    *y = 13;

    y = z;

    *y = 13;
}
```



???

y [0x234]

# 0x123

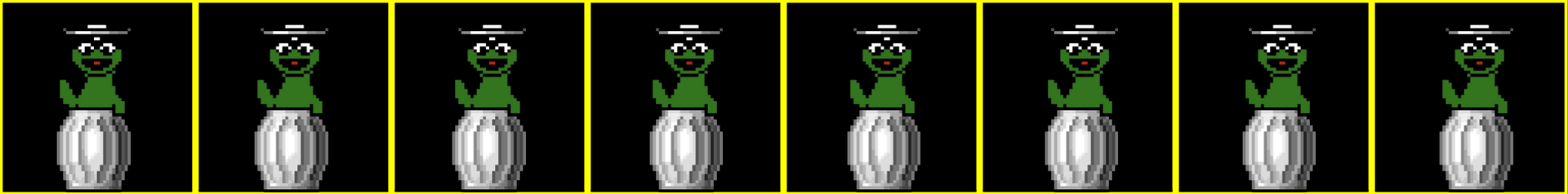
z [0x321]



7

x [0x123]





???

y [0x234]

0x123

z [0x321]



42

x [0x123]

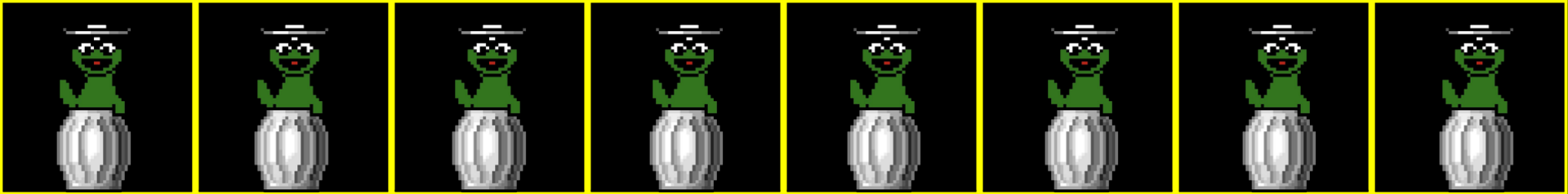


```
int main(void)
{
    int x = 7;
    int* y;
    int* z = &x;

    *z = 42;
    *y = 13;

    y = z;

    *y = 13;
}
```



???

y [0x234]

0x123

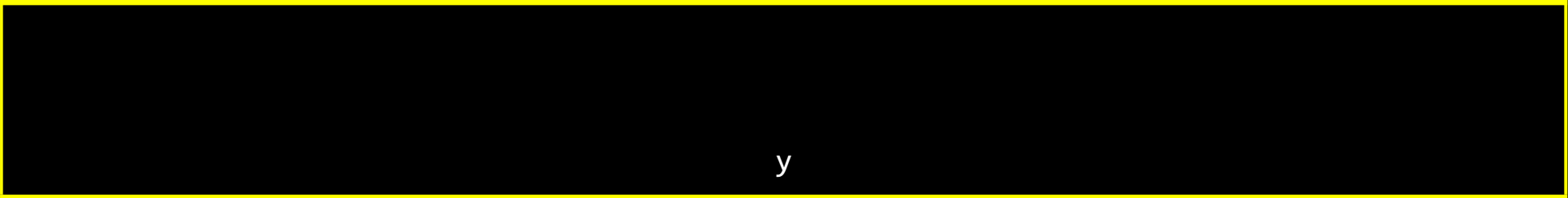
z [0x321]



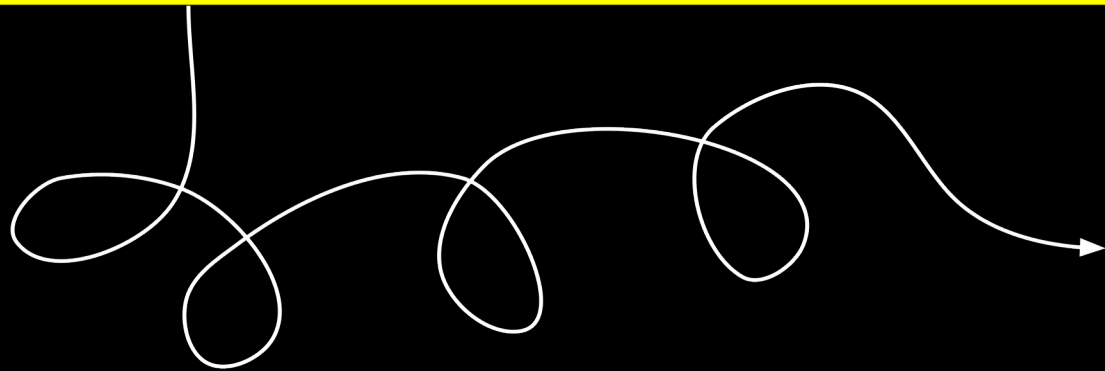
42

x [0x123]





y



```
int main(void)
{
    int x = 7;
    int* y;
    int* z = &x;

    *z = 42;
    *y = 13;

    y = z;

    *y = 13;
}
```

```
int main(void)
{
    int x = 7;
    int* y;
    int* z = &x;

    *z = 42;

    y = z;

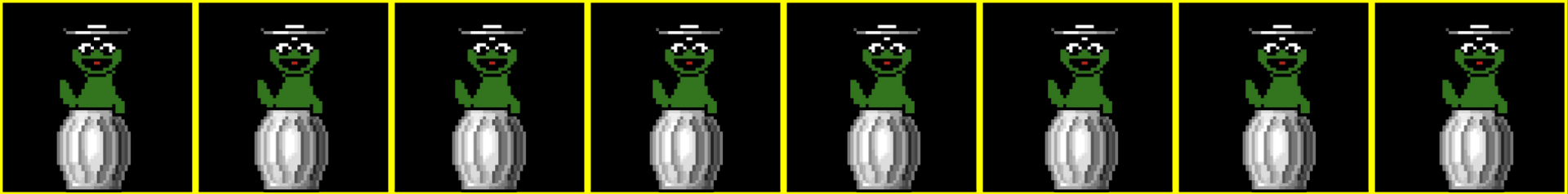
    *y = 13;
}
```

```
int main(void)
{
    int x = 7;
    int* y;
    int* z = &x;

    *z = 42;

    y = z;

    *y = 13;
}
```



???

y [0x234]

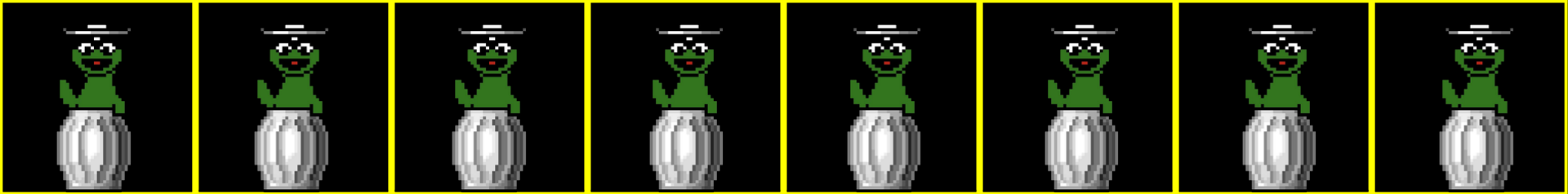
0x123

z [0x321]



42

x [0x123]



0x123  
y [0x234]

0x123  
z [0x321]



42  
x [0x123]

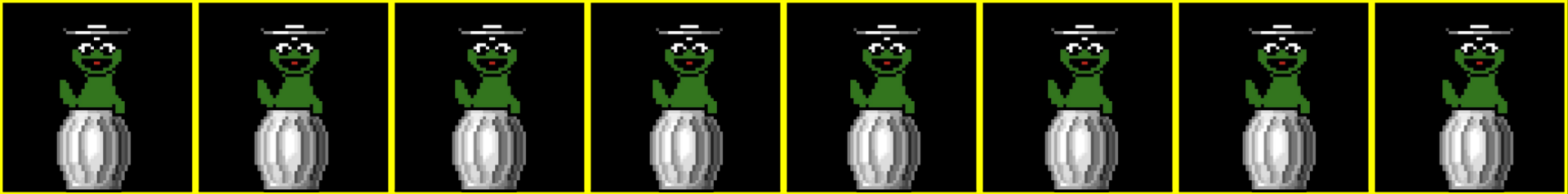


```
int main(void)
{
    int x = 7;
    int* y;
    int* z = &x;

    *z = 42;

    y = z;

    *y = 13;
}
```



0x123  
y [0x234]

0x123  
z [0x321]





0x123  
y [0x234]

0x123  
z [0x321]



13

x [0x123]



```
void trocar(int a, int b)
```

```
{
```

```
}
```

```
void trocar(int a, int b)
{
    int tmp = a;
    a = b;
    b = tmp;
}
```

```
void trocar(int a, int b)
```

```
{
```

```
    int tmp = a;
```

```
    a = b;
```

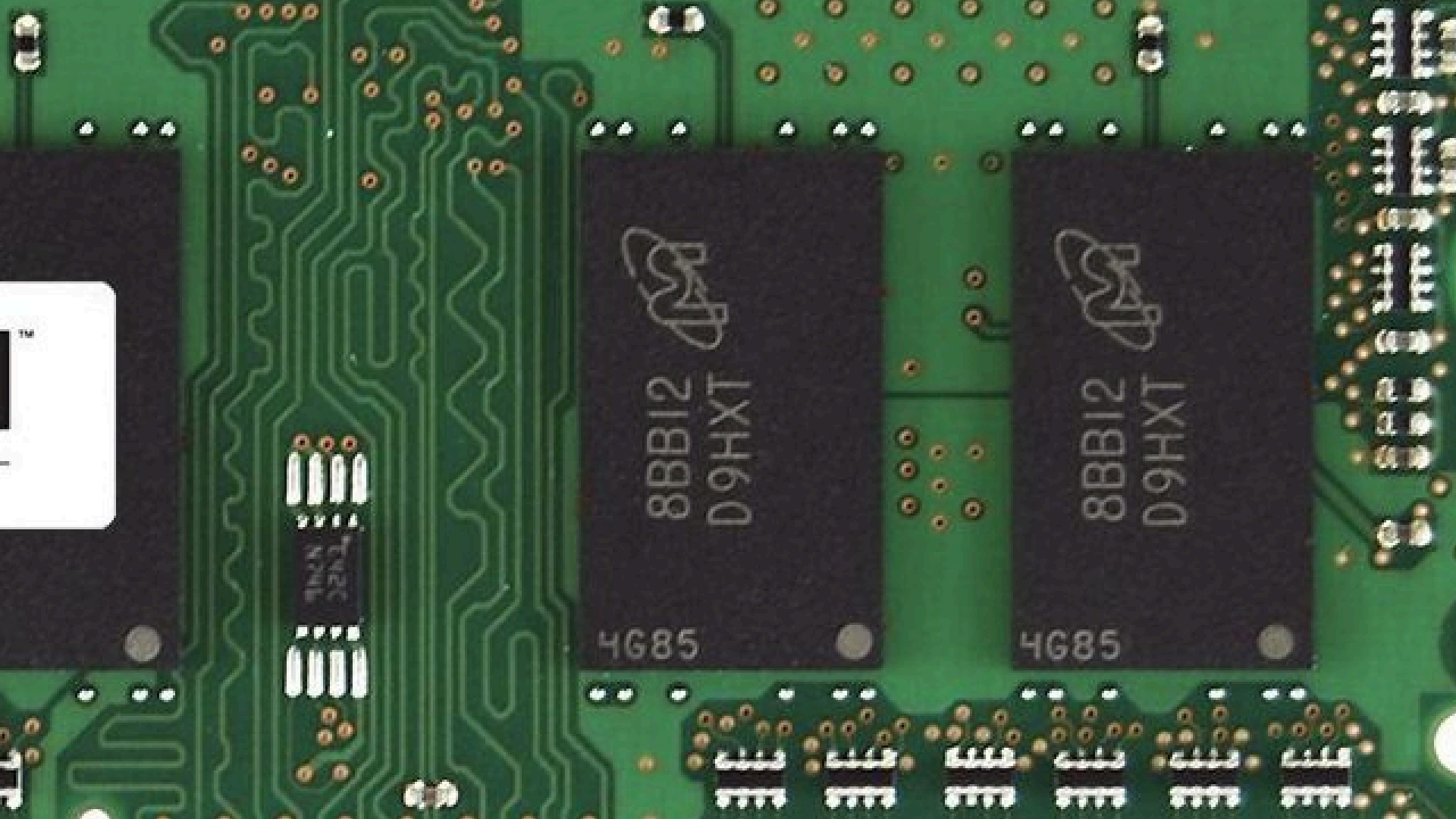
```
    b = tmp;
```

```
}
```

escopo

passagem por valor





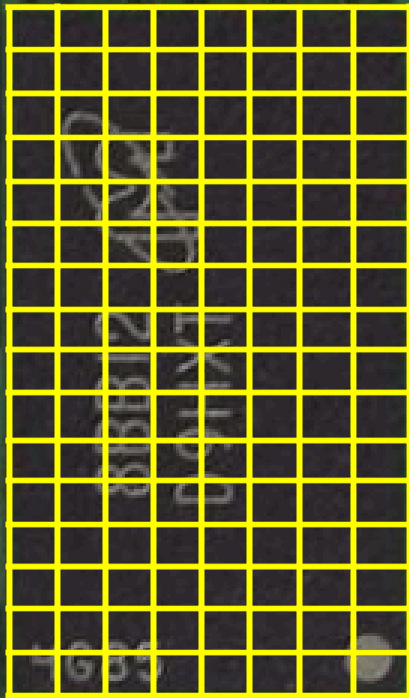
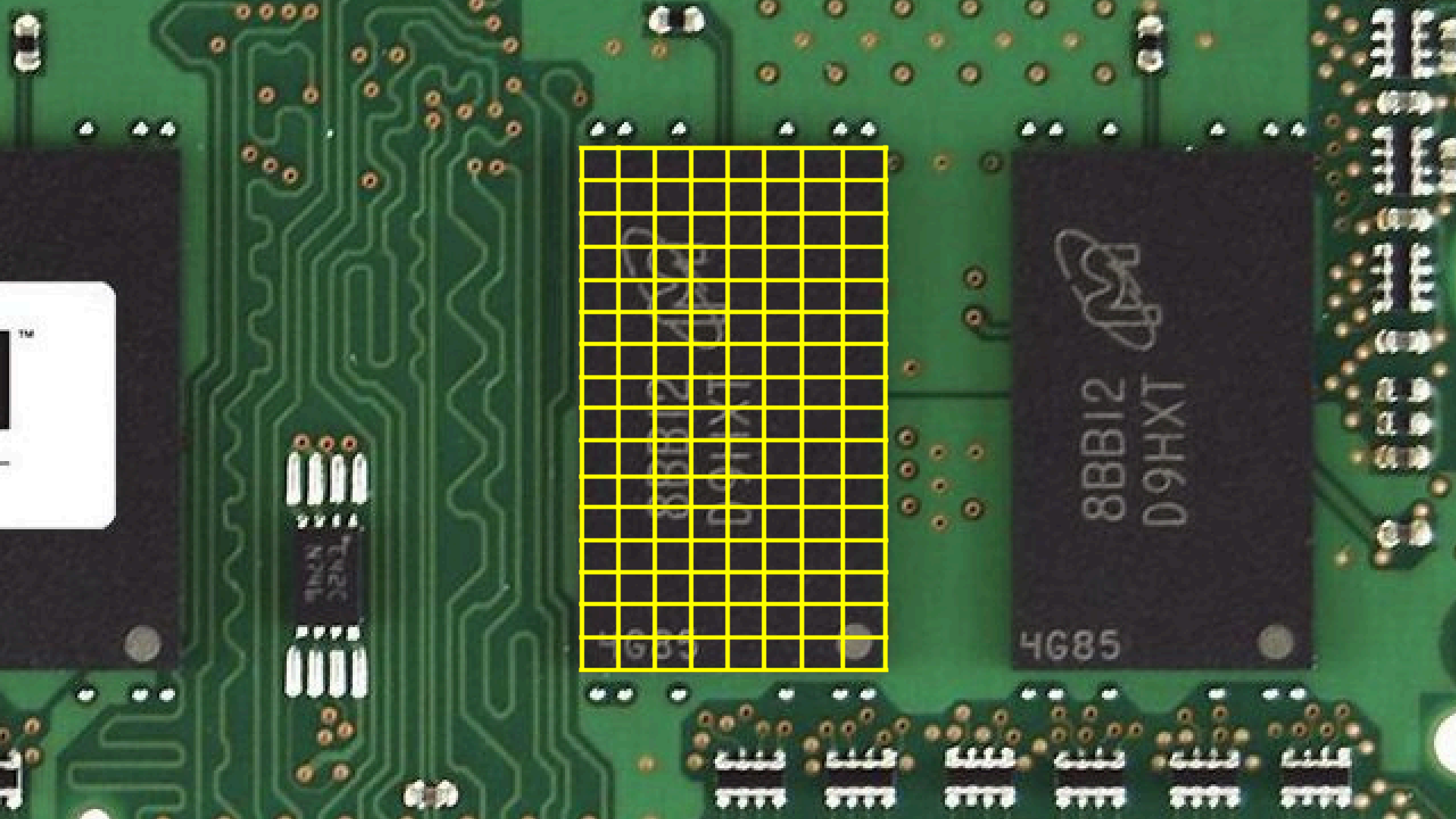
8BB12  
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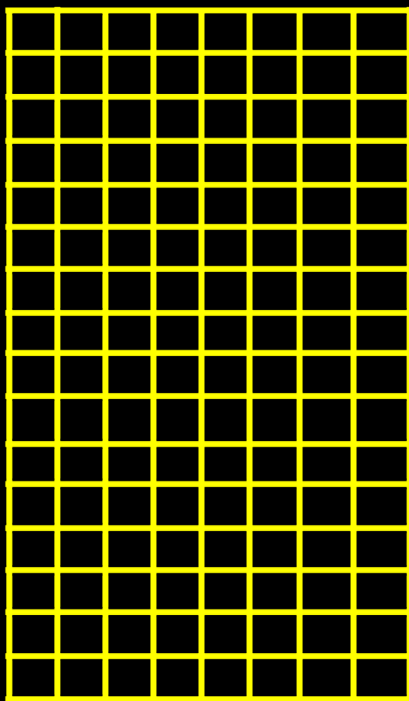
4G85

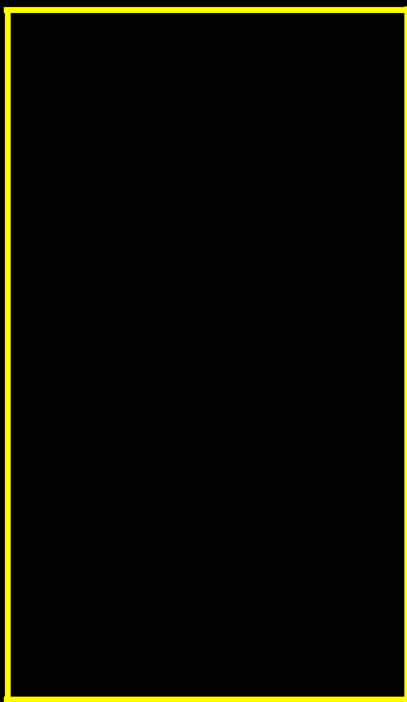
8BB12  
D9HXT

4G85

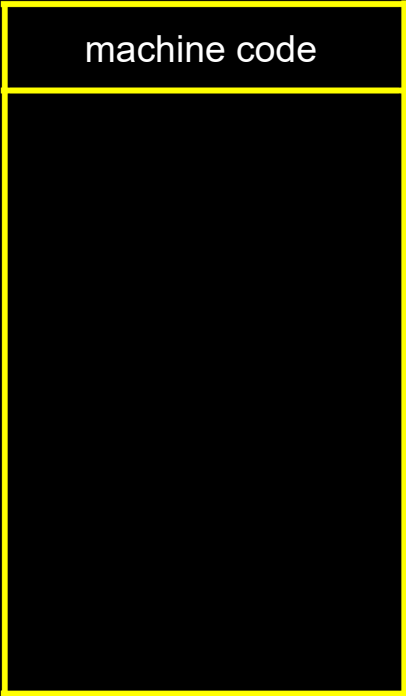
™





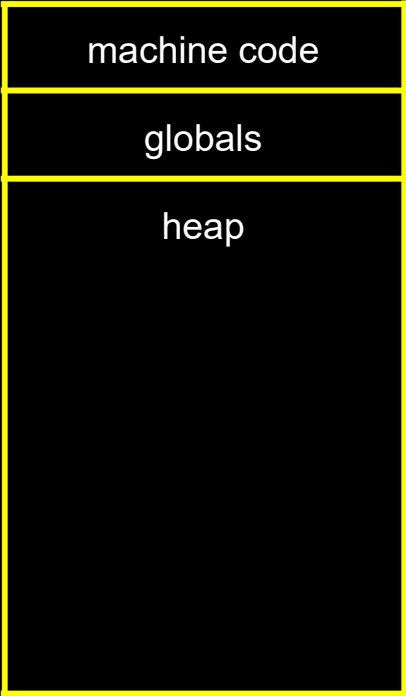


machine code

A diagram of a memory block. It consists of a yellow rectangular border. The top portion of the rectangle is filled with the text 'machine code' in white. The bottom portion of the rectangle is empty.

machine code

globals



A diagram showing a vertical stack of three memory sections. The top section is labeled 'machine code', the middle section is labeled 'globals', and the bottom section is labeled 'heap'. The sections are separated by horizontal lines, and the entire stack is enclosed in a yellow border.

|              |
|--------------|
| machine code |
| globals      |
| heap         |

machine code

globals

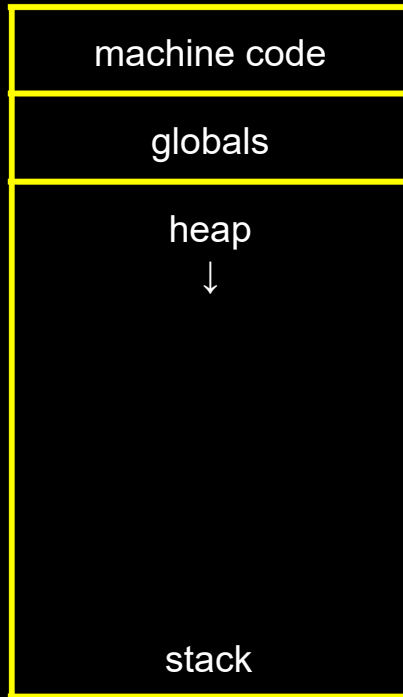
heap

machine code

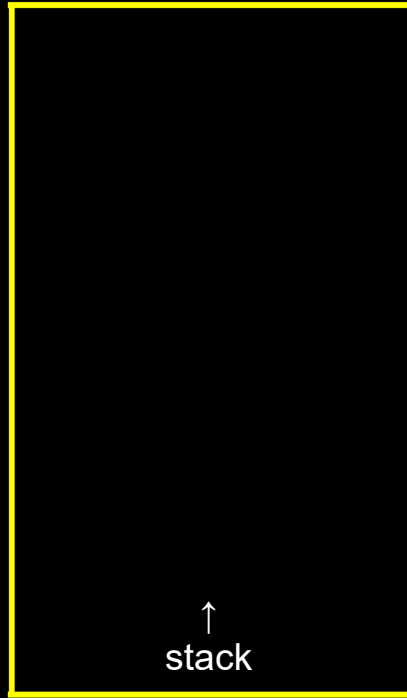
globals

heap









```
void trocar(int a, int b)
```

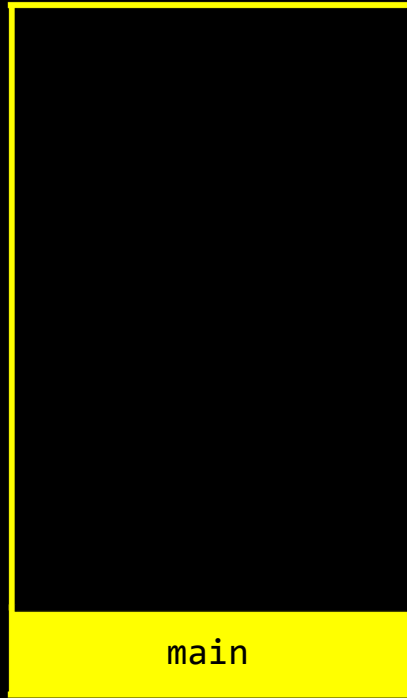
```
{
```

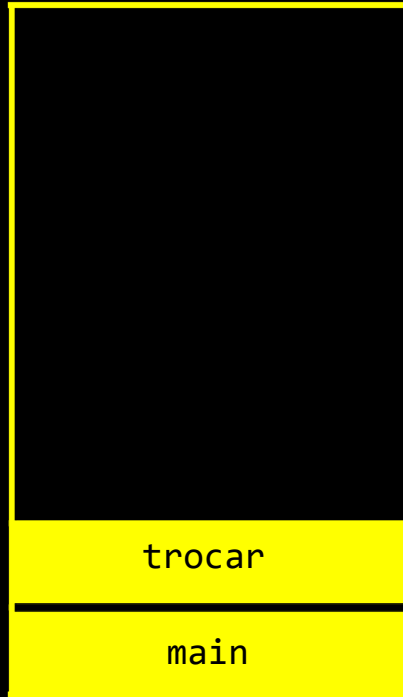
```
    int tmp = a;
```

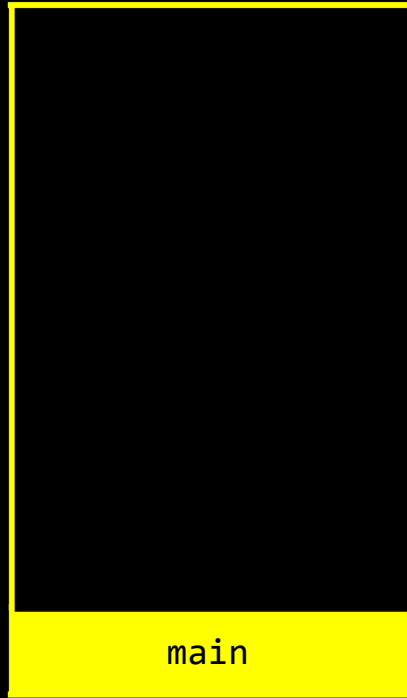
```
    a = b;
```

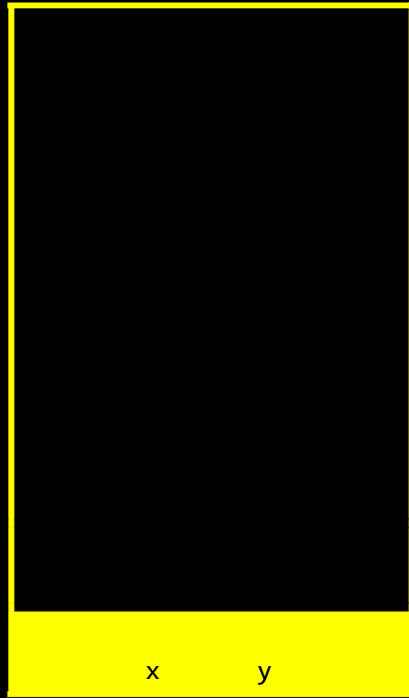
```
    b = tmp;
```

```
}
```

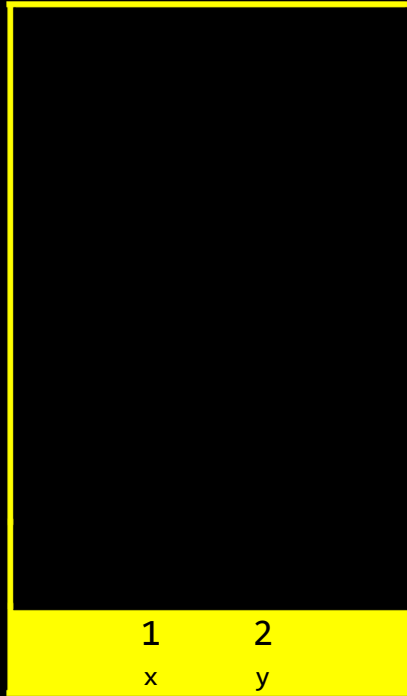








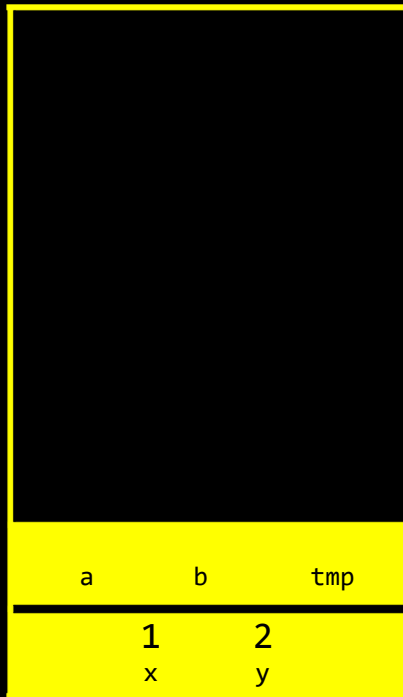
```
int x = 1;  
int y = 2;
```



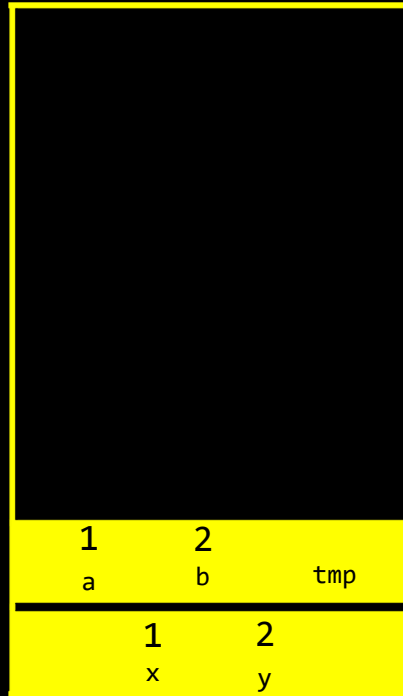
```
trocar(x, y);
```



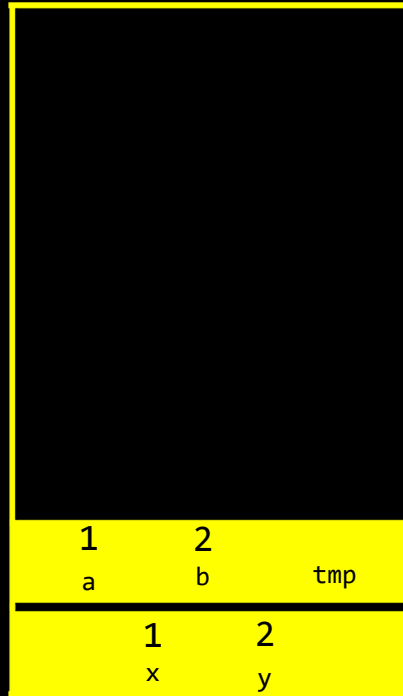
```
trocar(x, y);
```



```
void trocar(int a, int b);
```



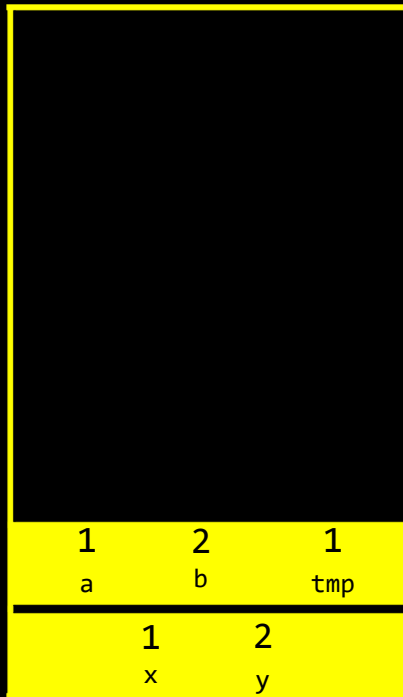
```
int tmp = a;  
a = b;  
b = tmp;
```



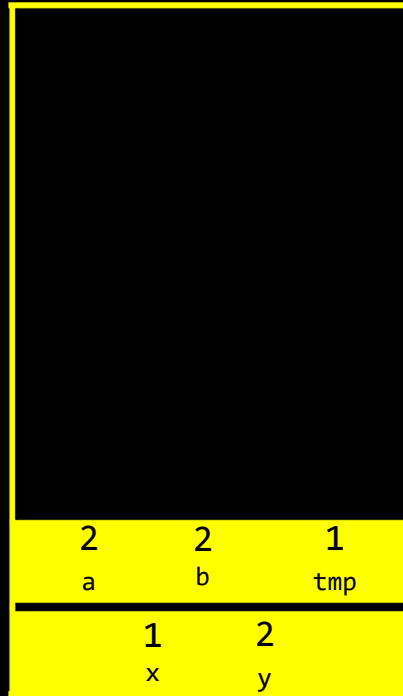
```
int tmp = a;
```

```
a = b;
```

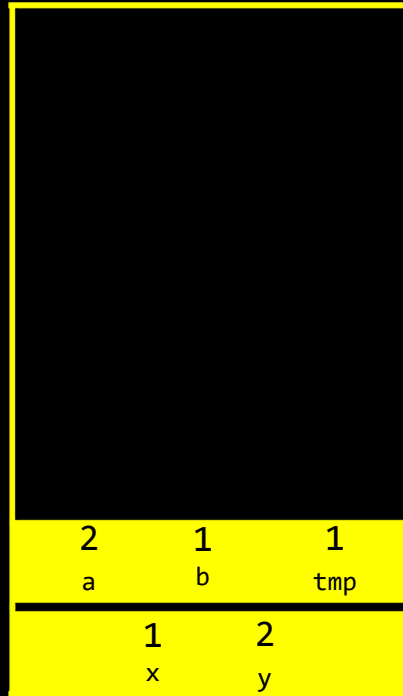
```
b = tmp;
```

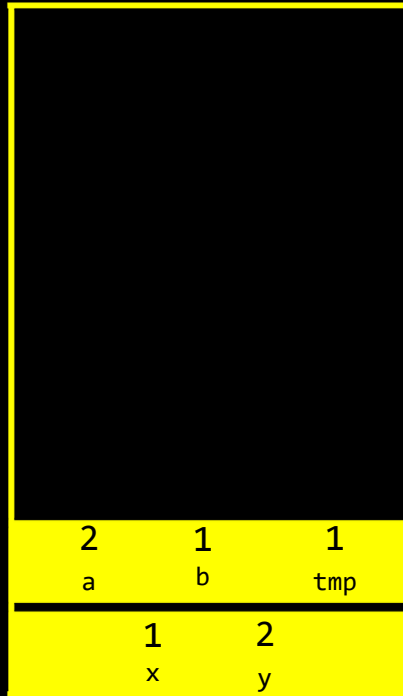


```
int tmp = a;  
a = b;  
b = tmp;
```



```
int tmp = a;  
a = b;  
b = tmp;
```





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|---|---|
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|   |   |
|   |   |
|   |   |
|   |   |
| 1 | 2 |
| x | y |

passagem por referência

```
void trocar(int a, int b)
```

```
{
```

```
    int tmp = a;
```

```
    a = b;
```

```
    b = tmp;
```

```
}
```

```
void trocar(int *a, int *b)
```

```
{
```

```
    int tmp = *a;
```

```
    *a = *b;
```

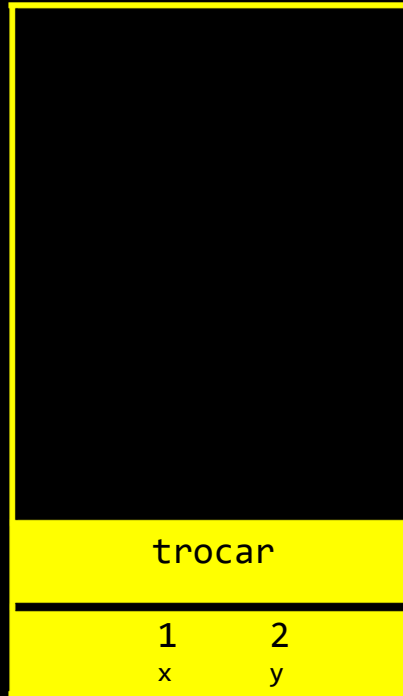
```
    *b = tmp;
```

```
}
```

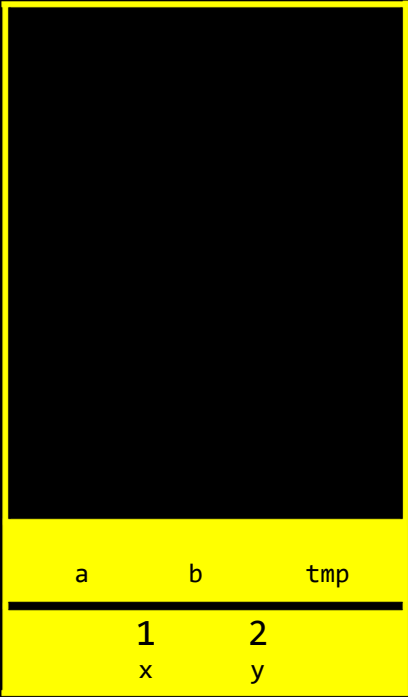
```
void trocar(int *a, int *b)
{
    int tmp = *a;
    *a = *b;
    *b = tmp;
}
```

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|---|---|
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|   |   |
|   |   |
|   |   |
|   |   |
|   |   |
| 1 | 2 |
| x | y |

```
trocar(x, y);
```



```
trocar(x, y);
```



```
void trocar(int *a, int *b);
```



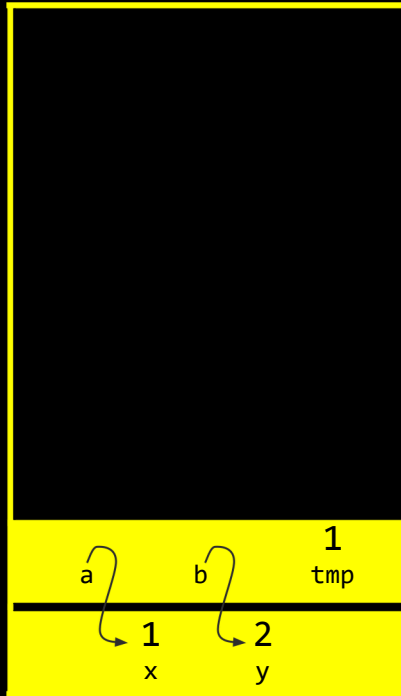
```
int tmp = *a;  
*a = *b;  
*b = tmp;
```



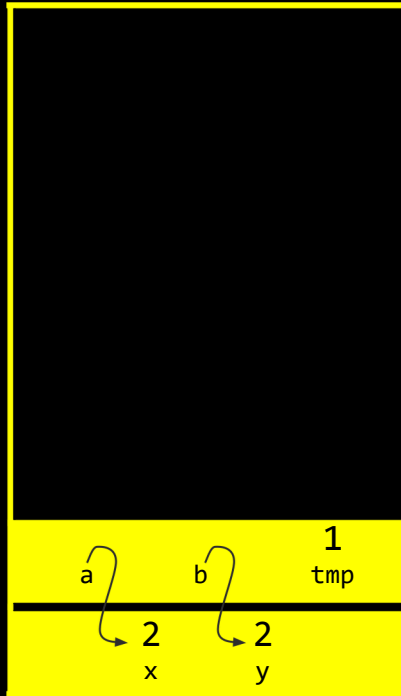
```
int tmp = *a;
```

```
*a = *b;
```

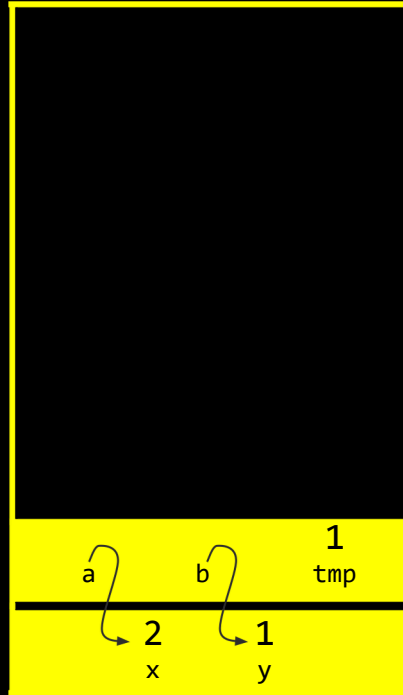
```
*b = tmp;
```

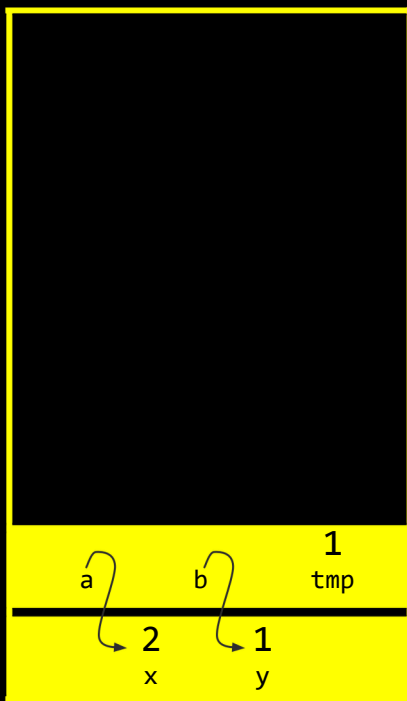


```
int tmp = *a;  
*a = *b;  
*b = tmp;
```



```
int tmp = *a;  
*a = *b;  
*b = tmp;
```





|   |   |
|---|---|
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|   |   |
|   |   |
|   |   |
|   |   |
|   |   |
|   |   |
|   |   |
| 2 | 1 |
| x | y |

